

## Assessment test-3

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Artificial Intelligence

### Introduction:-

A smart home automation system is being designed to manage lighting, temperature, and security based on user behaviour and environmental condition. A smart home receives input from the sensor, process the information, and output.

There are four main types of intelligences

1. Simple Reflex Agent
2. Model Based Reflex Agent
3. Goal Based Agent
4. Utility Based Agent.

#### 1. Simple Reflex Agent:-

A simple reflex Agent makes decision based only on the current input from the environment. It follows predefined condition action rules and does not remember past events.

## Uniform cost search:-

- Expands the path with the lowest cumulative cost first
- Uses a priority queue
- Find the minimum - cost path

Ex:- if one path costs 8 and another 5 UCS chooses the path with cost 5

## Advantages:-

- Complete
- Finds the optimal path

## Disadvantages:-

- High memory usage
- Slow for large search spaces

## Iterative Deepening Search:-

- Performs depth-limited search repeatedly
- Increases search depth one level at a time
- Combines advantages of DFS and BFS

## Advantages:-

- Low memory usage

### Disadvantages:

- Requires planning algorithms
- Slower than reflex agents.

## 4. Utility-Based Agent:

A Utility Based agent selects the action that provides the maximum overall benefit by considering multiple factors such as comfort, energy consumption, and cost.

### Working:-

- Evaluates all possible actions
- Calculates the utility value for each action
- Chooses the action with highest utility

### Advantages:-

- Produces optimal decision
- Learn user preferences

### Disadvantages:-

- Utility calculation is complex
- More expensive to implement

## 2. Introduction:-

A robot placed in an unknown maze must explore different paths to find the exit.

The important uninformed search algorithms are Uniform Cost Search and Iterative Deepening Search.

## 2. Model Based Reflex agent :-

A model based Reflex agent maintains an internal model of the environment. It combines current sensor information with previous observation to make better decisions.

### Working:-

- Receives sensor input
- Update its internal model
- Uses current and past information to decide the action

### Advantages:-

- Can handle partially observable environments
- Uses memory to improve accuracy

### Disadvantages:-

- Requires more memory
- More complex to design

## 3. Goal - Based Agent

A Goal Based Agent chooses action that help achieve a specific goal. It evaluates different possibilities before making a decision.

### Working:-

- Defines a goal
- Examines possible action
- Selects the best action that reaches the goal

### Advantages:-

- Flexible and intelligent



o Suitable when goal depth is unknown

Disadvantages :-

o Repeats some searches

o Not optimal when path costs differ